

CSE 311:

Data Communication

Instructor:
Dr. Md. Monirul Islam

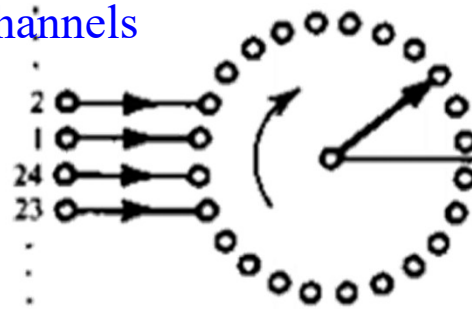
Sampling and Analog-to-Digital Conversion

PCM in TDM: T1 Carrier System

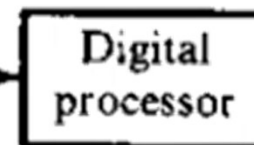
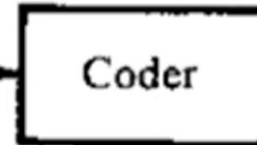
- A Case Study

Review

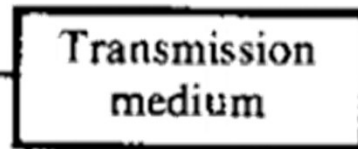
24 input
channels



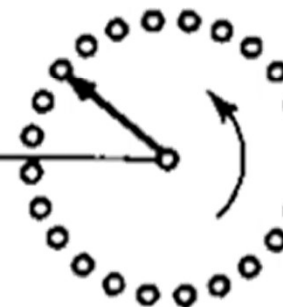
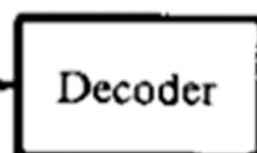
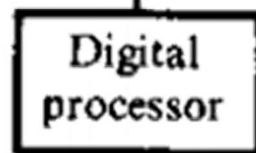
Commutators



DS1 (1.544 Mbps)

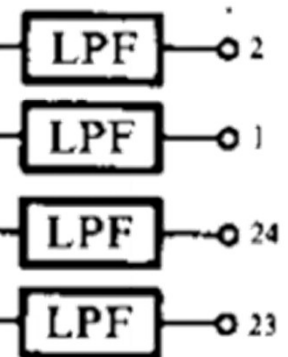


with regenerative repeaters,
each one is 6K feet apart



Decommutators

24 output
channels

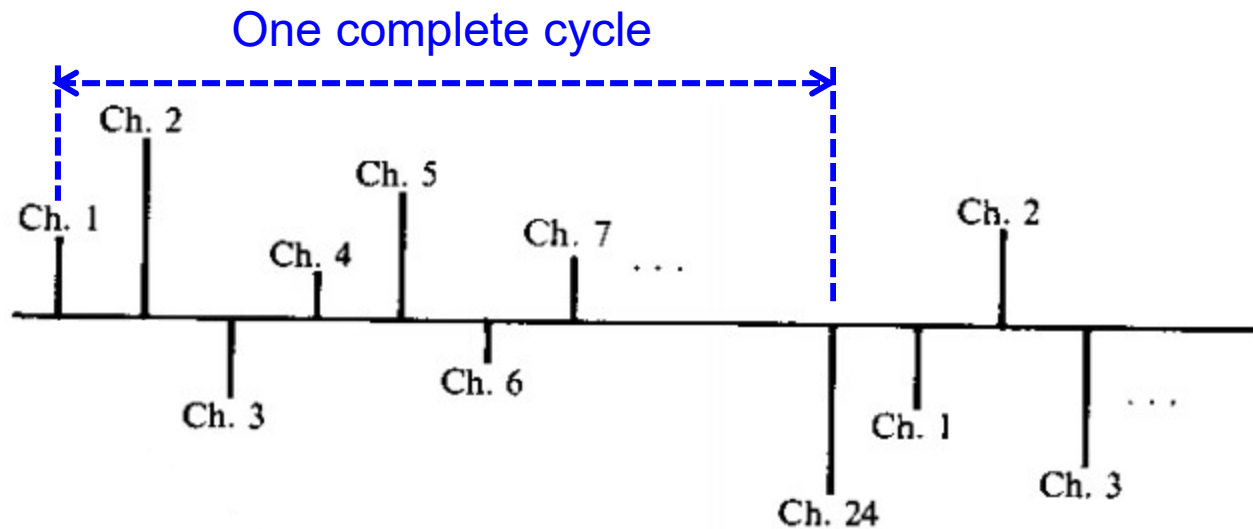
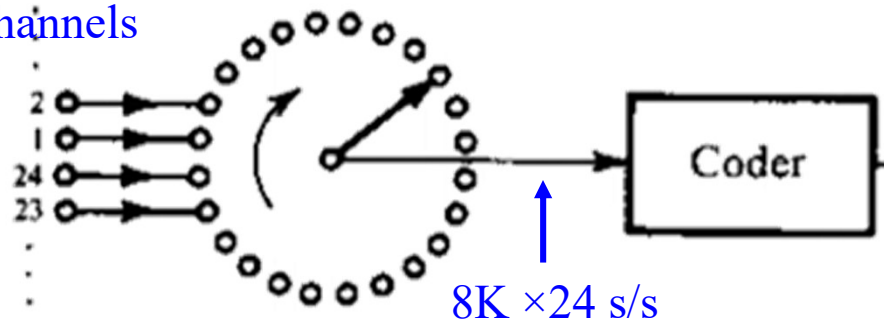


PCM in TDM: T1 Carrier System

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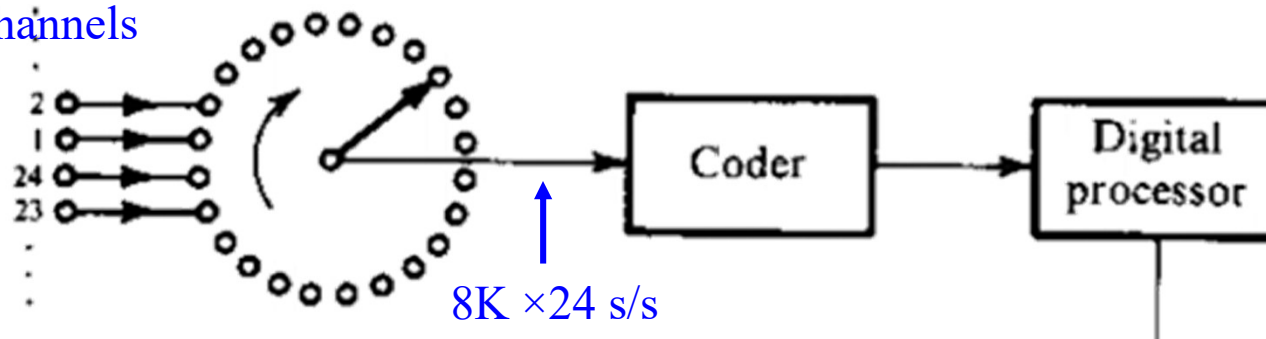


PCM in TDM: T1 Carrier System

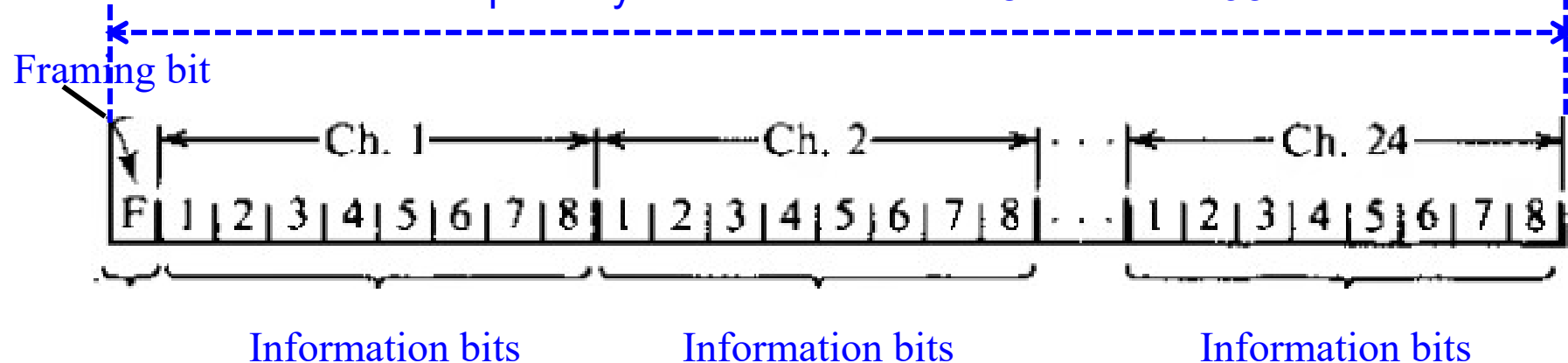
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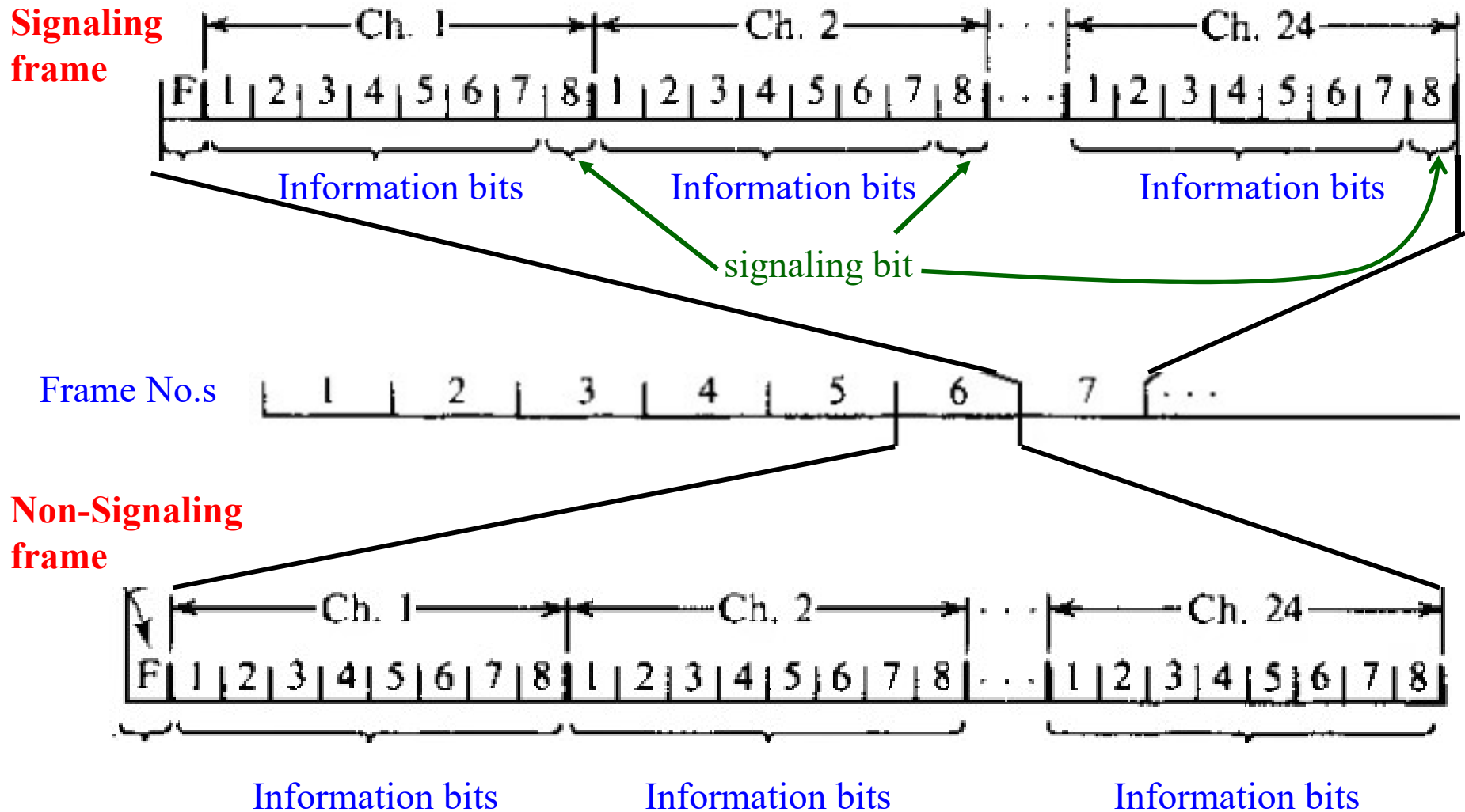
24 input
channels



1 complete cycle = 1 Frame = $1 + 8 \times 24 \text{ bits} = 193 \text{ bits}$

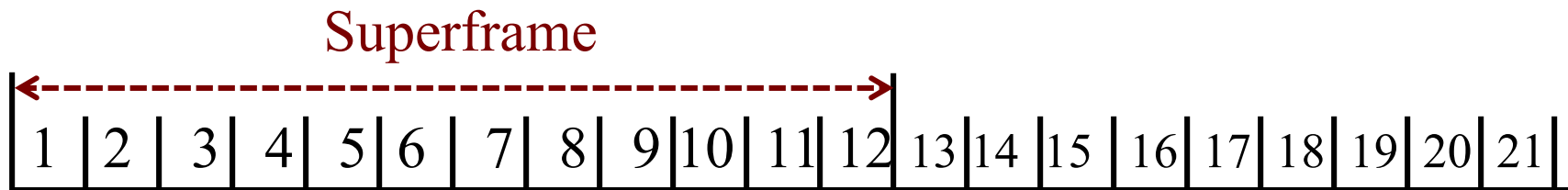


How to Differentiate Signaling and Non-Signaling Frames?



How to Differentiate Signaling and Non-Signaling Frames?

Concept of Superframe

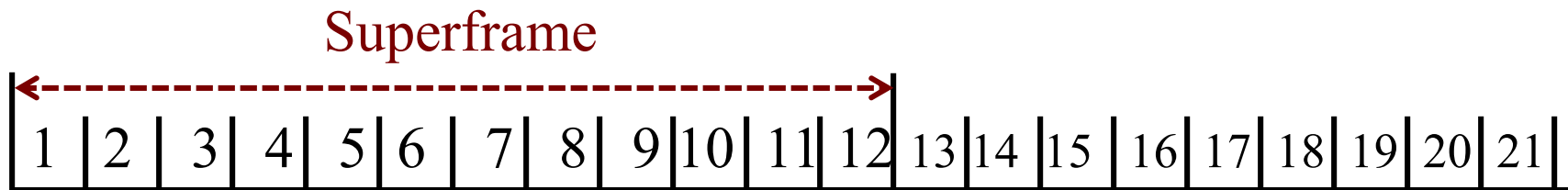


Frame No.s

- 12 frames make a Superframe
- Framing bits form a special pattern 100111011100

How to Differentiate Signaling and Non-Signaling Frames?

Concept of Superframe

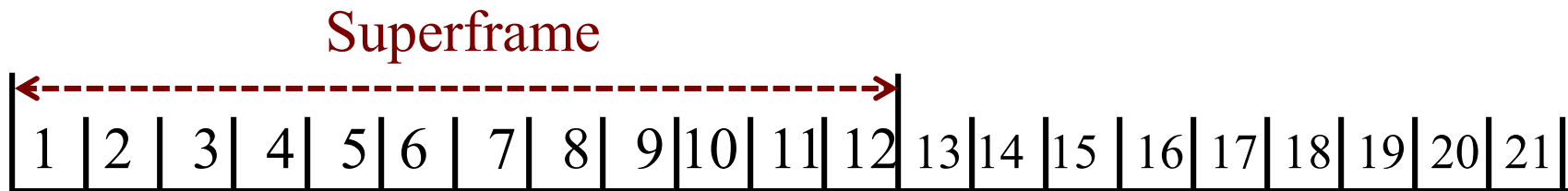


Frame No.s

- The special pattern **100111011100** repeats in every 12 frames
- Identification of 6th and 12th frame is easy, and so is signaling frames.

How to Differentiate Signaling and Non-Signaling Frames?

Concept of Superframe



Frame No.s

- 2 signaling frames in a superframe
- 2 signaling bits in a superframe
- 4 state signaling is possible

Extended Superframe (ESF)

Structure of ESF



Frame No.s

- 24 frames make an ESF
- Reduces framing bits
- Signaling bits in the LSB of 6th, 12th, 18th and 24th frames
- 4 signaling bits makes possibility of 16 state-signaling

Extended Superframe (ESF)

Structure of ESF



Frame No.s

- Reduces framing bits: every 4th frame has a framing bit
- 8 Kbps framing capacity is distributed to
 - 2 Kbps for framing
 - 2 Kbps for CRC-6
 - 4 Kbps for data

Extended Superframe (ESF)

Structure of ESF



Frame No.s

- Reduces framing bits: every 4th frame has a framing bit
framing bits make a pattern **001011**: repeats in every 24 frames

Further improvement: Common Channel Interoffice Signaling

CCIS

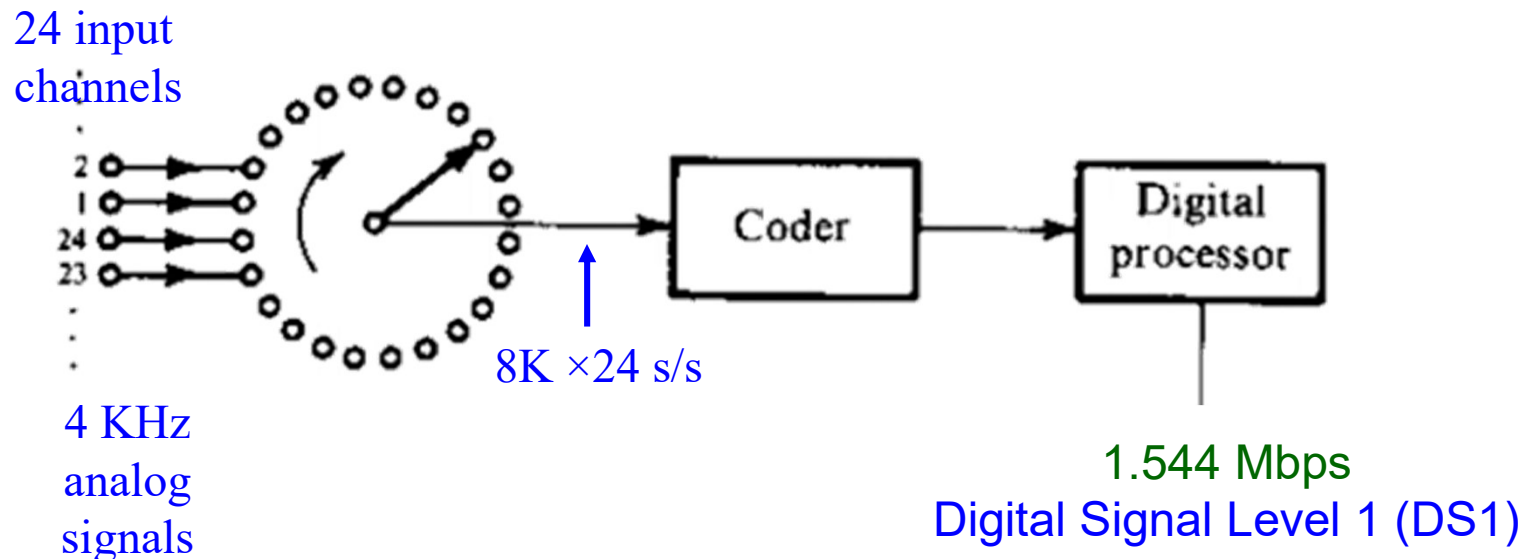
- A separate network for signaling
- decreases the necessity of robbed-bit-signaling
- All 8 bits can be used for transmitting information bits

European Counterpart of T1 System

Conference on European Postal and Telegraph Administration (CEPT)

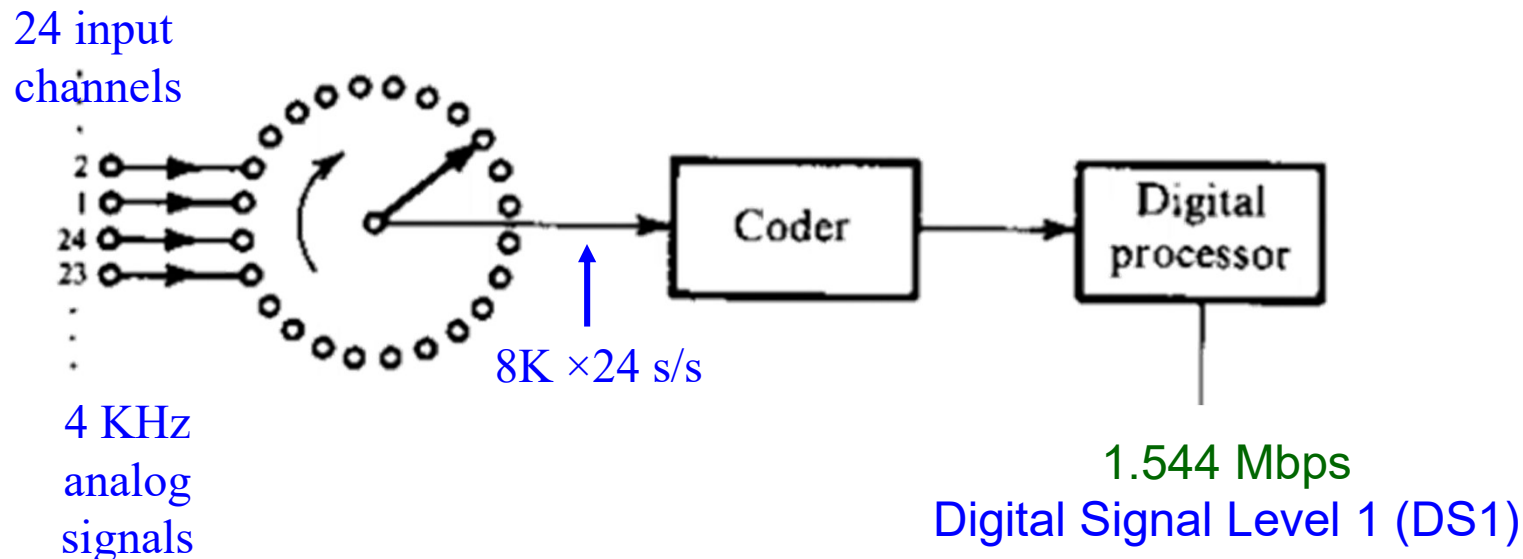
- standardizes PCM with 256 time slots per frame
- capable to transmit 32 channel audio signals simultaneously
- uses 30 channels with $30 \times 8 = 240$ information bits per frame
- uses remaining 16 bits for sync and signaling

Digital Multiplexing: Generalization of T1 System



- Converts 24 4-KHz analog signals to digital signal of 1.544 Mbps

Digital Multiplexing: Generalization of T1 System



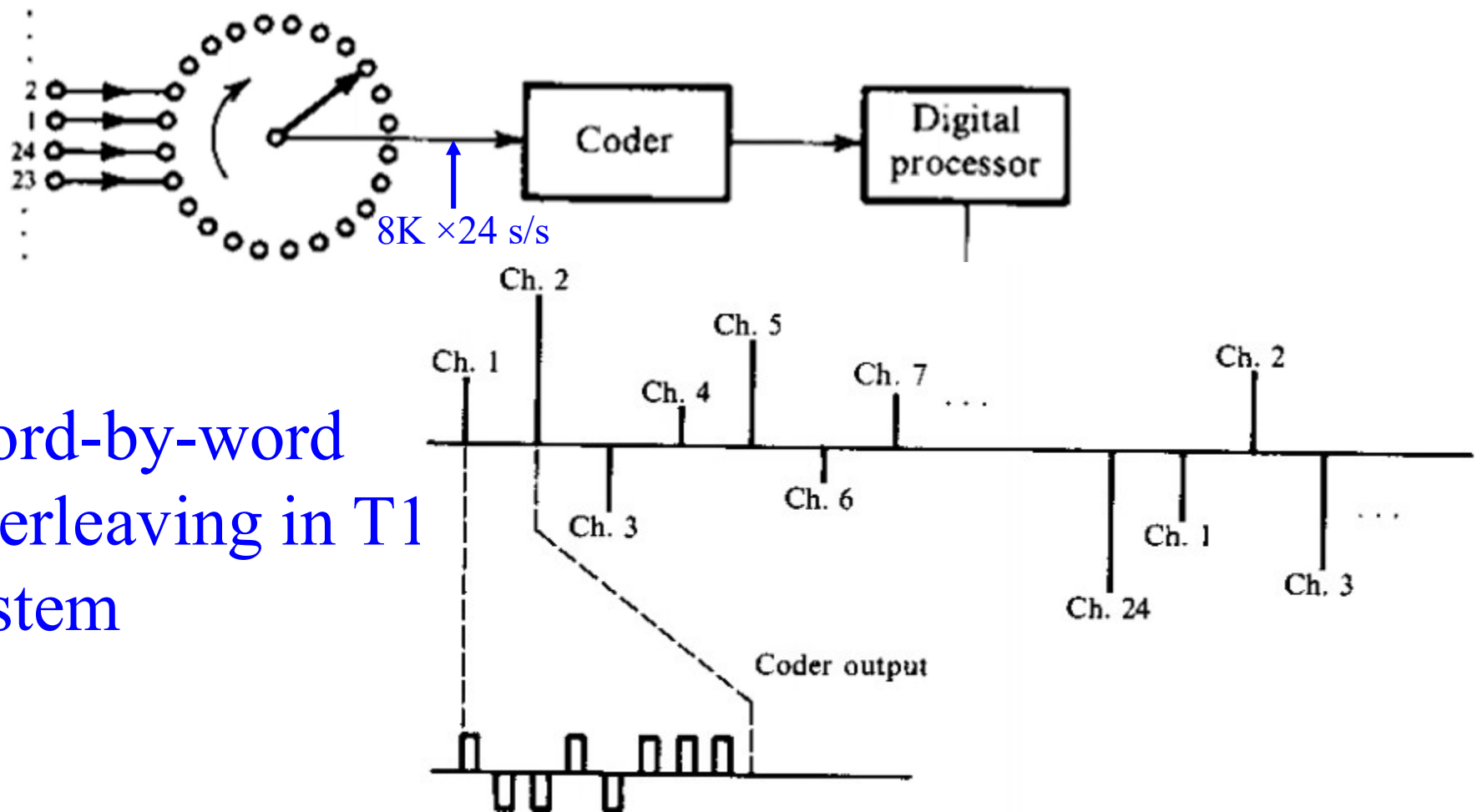
- Several **slow-bit-rate** digital signals can be combined to get high-bit-rate signal and can **time-share** same medium (**TDM**)
- Input bit rates may be **different**

Digital Multiplexing: Generalization of T1 System

TDM can be

- Word-by-word basis interleaving (like T1 system)
- Bit-by-bit basis interleaving

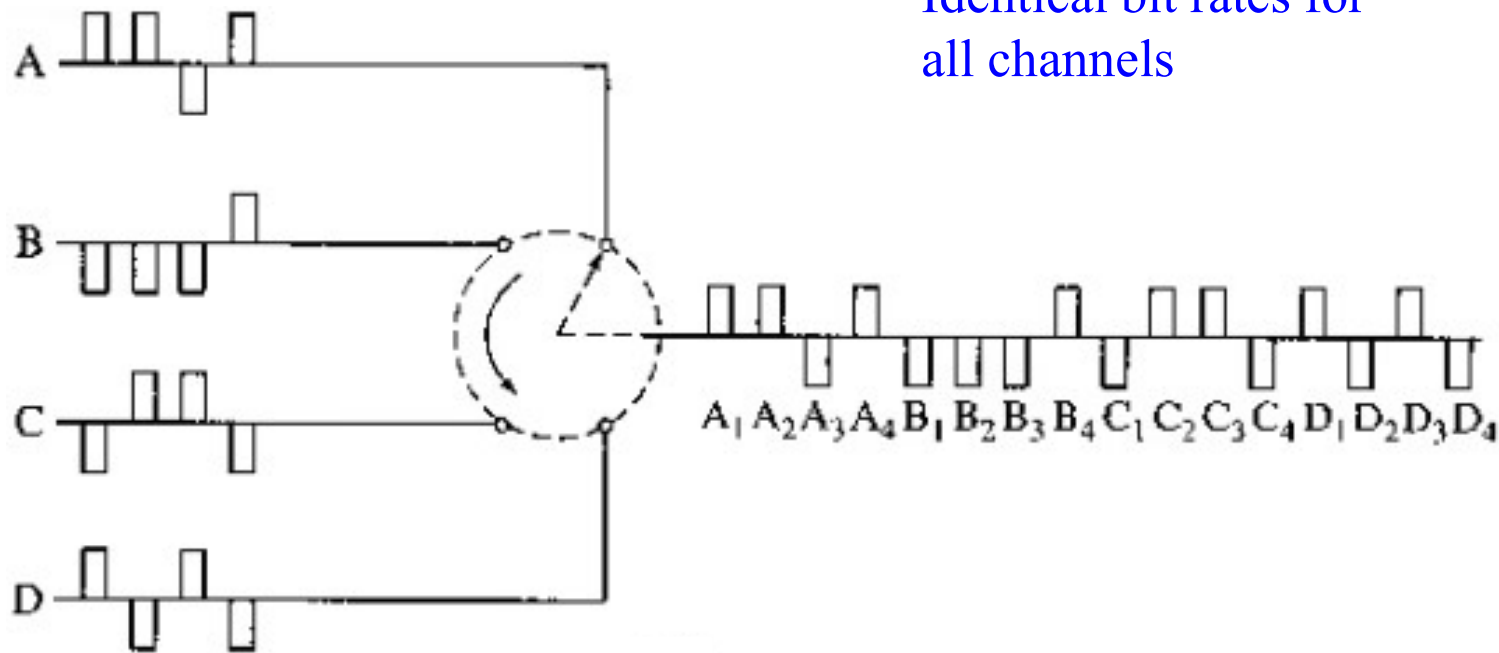
Digital Multiplexing: Generalization of T1 System



Word-by-word
interleaving in T1
system

Digital Multiplexing: Word Interleaving

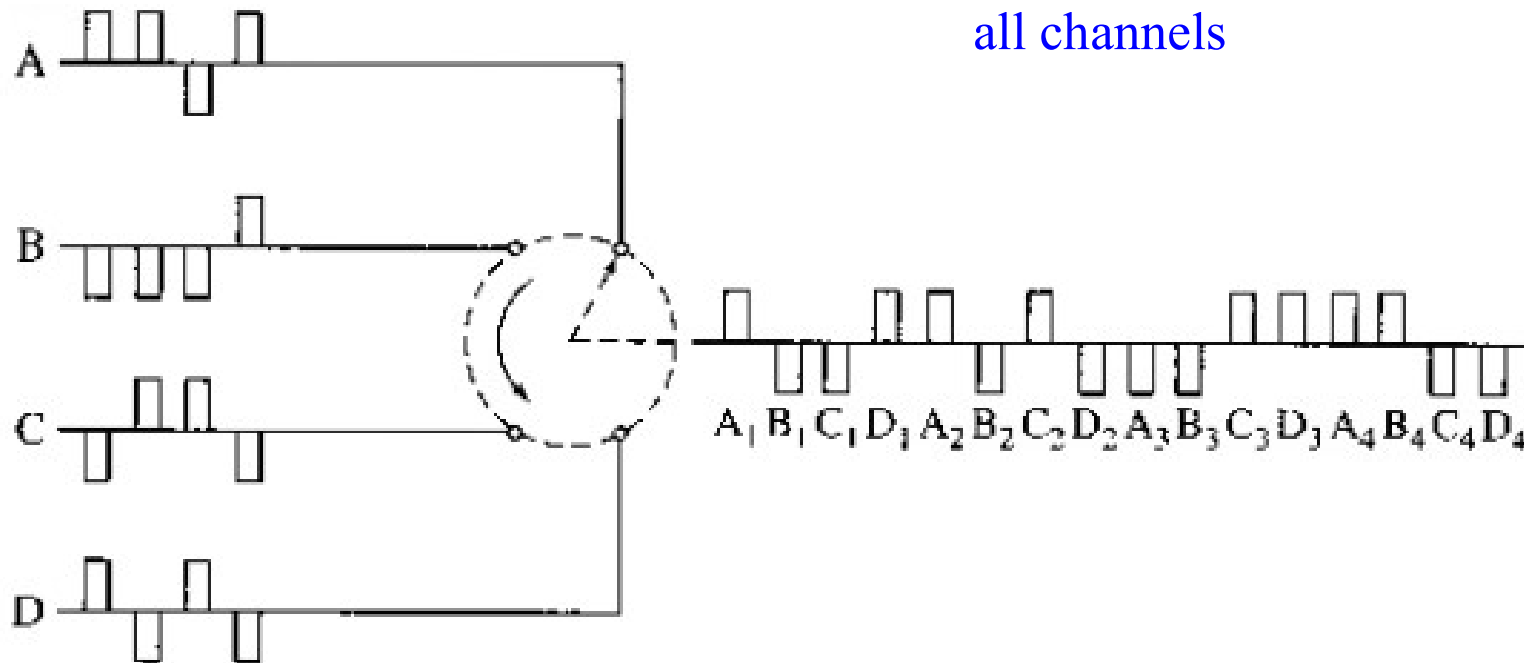
Channels



Used in: T1 system to generate DS1

Digital Multiplexing: Bit/Digit Interleaving

Channels

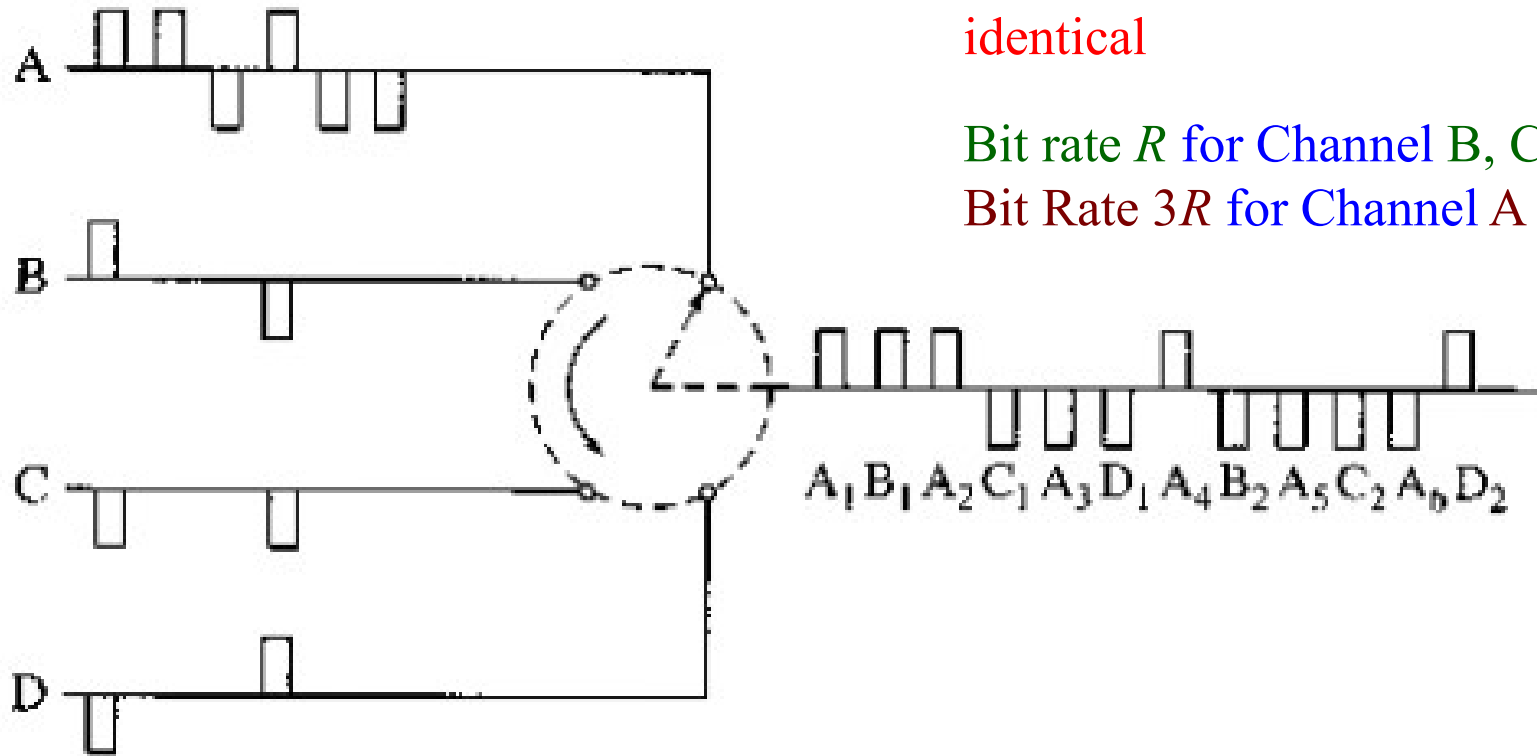


Identical bit rates for
all channels

Used in: North American Digital
Hierarchy

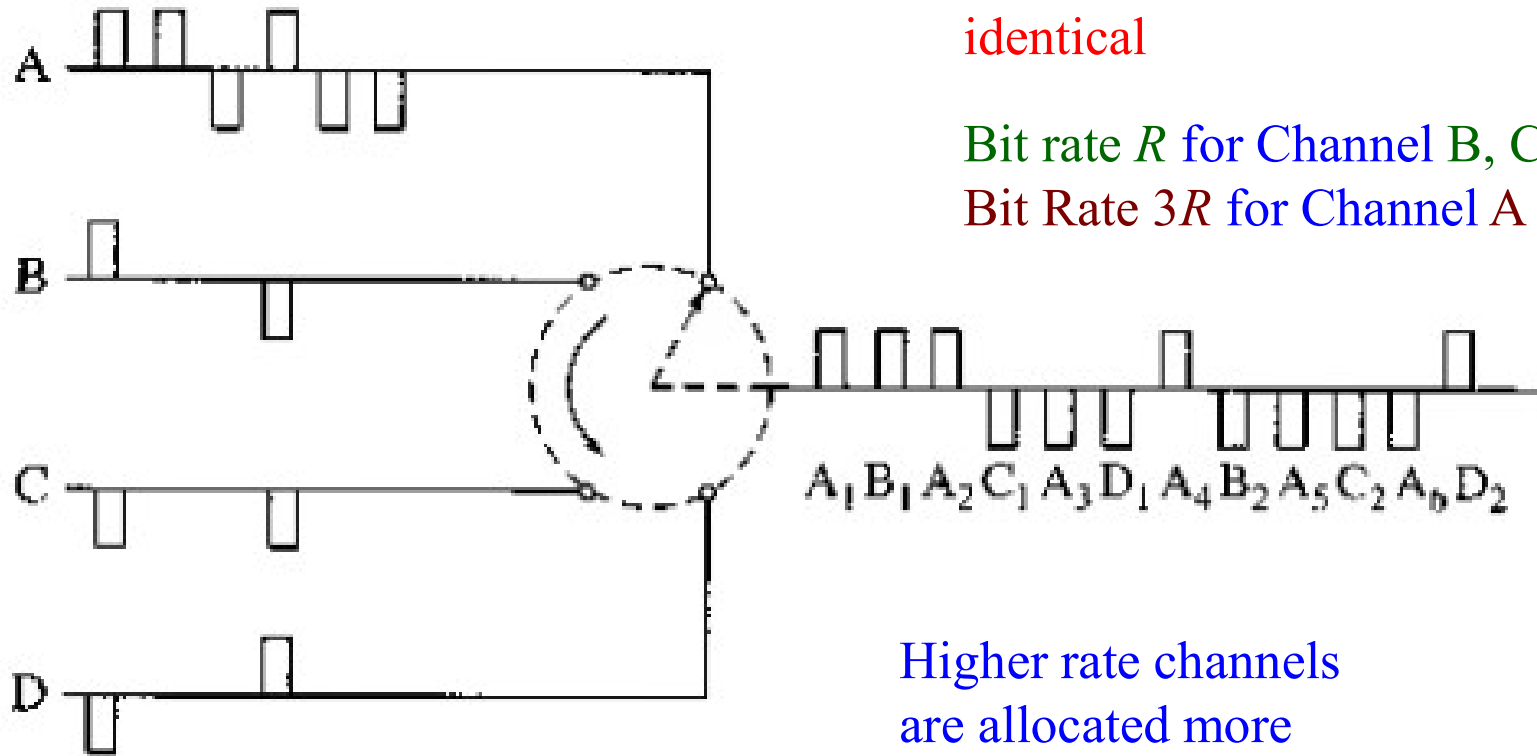
Digital Multiplexing: Bit/Digit Interleaving

Channels



Digital Multiplexing: Bit/Digit Interleaving

Channels



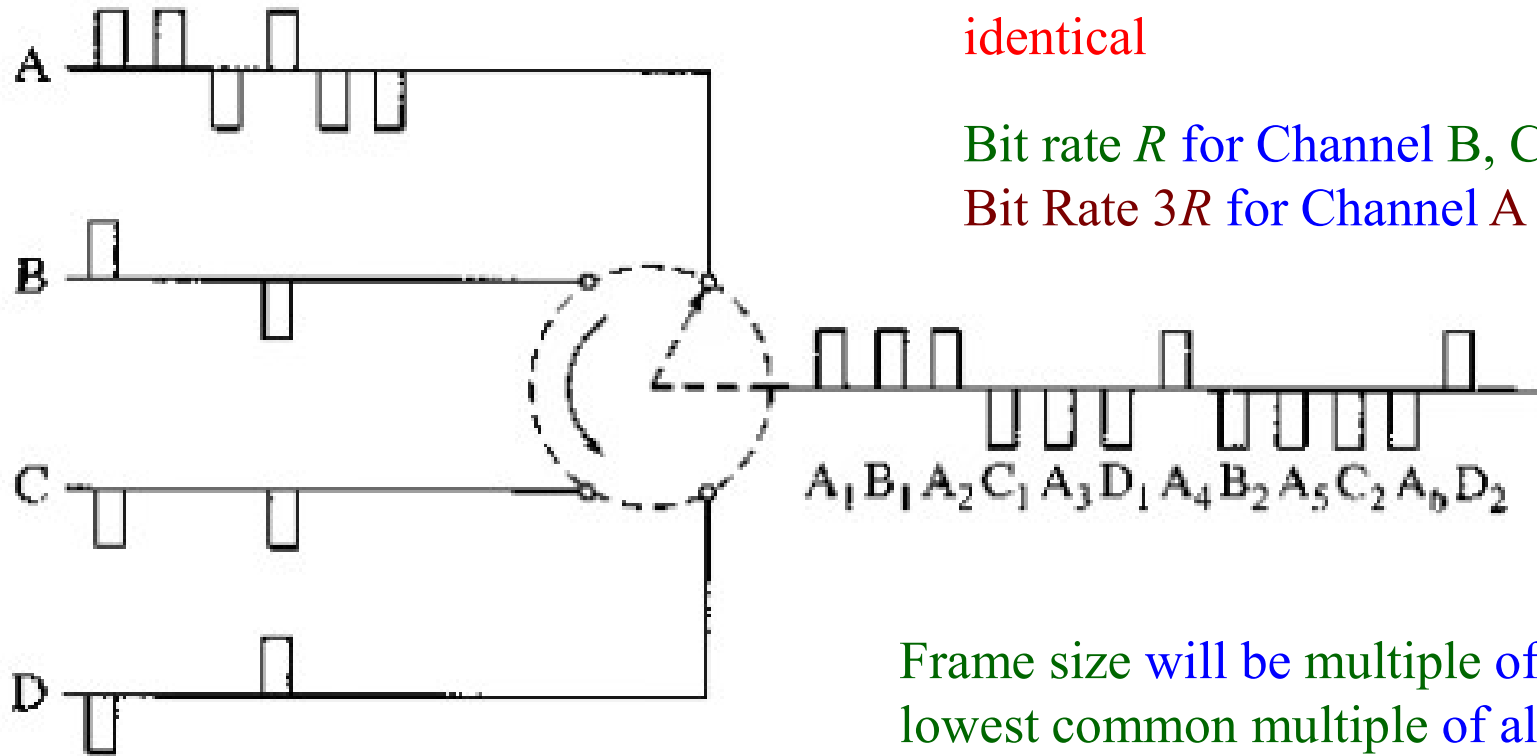
When bit rates are **not**
identical

Bit rate R for Channel B, C and D
Bit Rate $3R$ for Channel A

Higher rate channels
are allocated more
slots

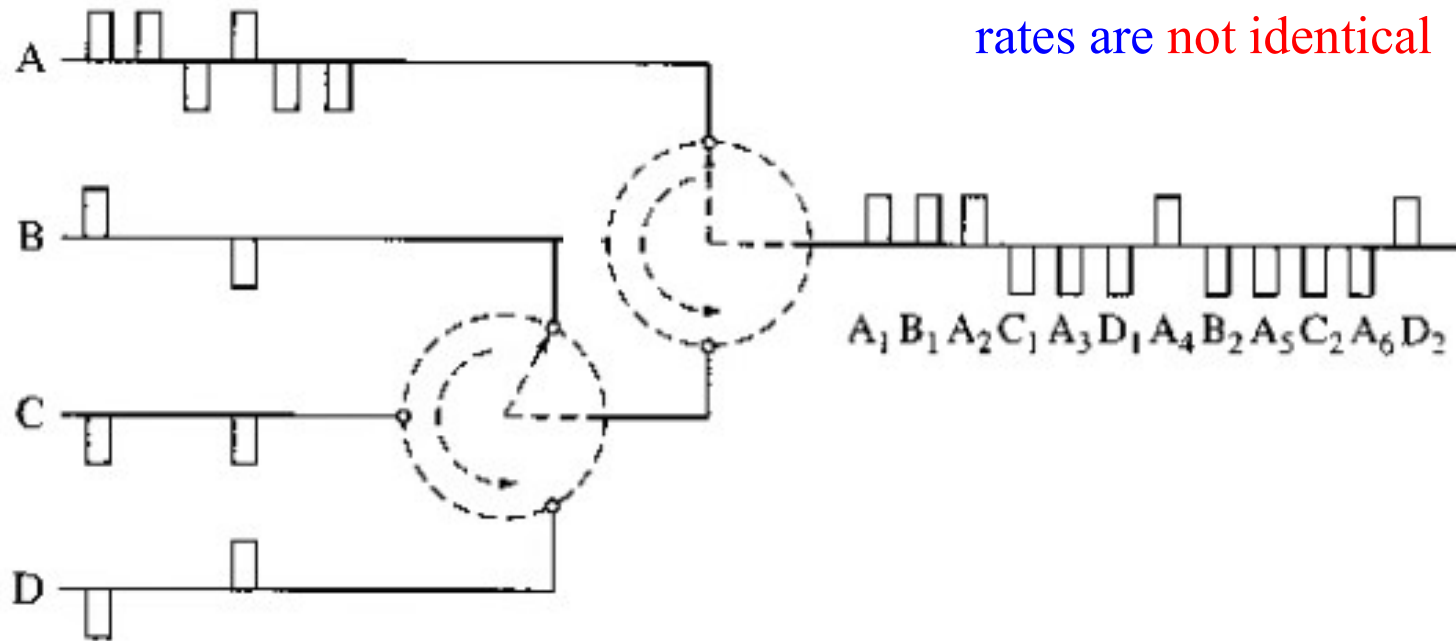
Digital Multiplexing: Bit/Digit Interleaving

Channels



Digital Multiplexing: Bit/Digit Interleaving

Channels



A different architecture when bit rates are not identical

Digital Multiplexing: Receiver Side

- Data has to be divided and distributed in different channels
- Requires correct identification of bits which is correct identification of
 - beginning of each frame
 - slots in each frame
 - bit in a slot

Digital Multiplexing: Receiver Side

- Data has to be divided and distributed in different channels
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All these are done by adding overhead bits which consist of

- Framing and synchronizing bits

Extended Superframe (ESF)

Structure of ESF



Frame Numbers

- Reduces framing bits: every 4th frame has a framing bit
framing bits make a pattern **001011**: repeats in every 24 frames

DM1/2 Multiplexer Frame Format

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
M_1	[48]	C_B	[48]	F_0	[48]	C_B	[48]	C_B	[48]	F_1	[48]
M_1	[48]	C_C	[48]	F_0	[48]	C_C	[48]	C_C	[48]	F_1	[48]
M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

- DM1/2: converts $4 \times \text{DS1}$ (1.544 Mbps) to $1 \times \text{DS2}$ (6.312 Mbps)
- Bit-by-bit interleaving of 4 channels of 1.544 Mbps
- The frame has 4 sub-frames
- Each sub-frame has 6 overhead bits

DM1/2 Multiplexer Frame Format

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
M_1	[48]	C_B	[48]	F_0	[48]	C_B	[48]	C_B	[48]	F_1	[48]
M_1	[48]	C_C	[48]	F_0	[48]	C_C	[48]	C_C	[48]	F_1	[48]
M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

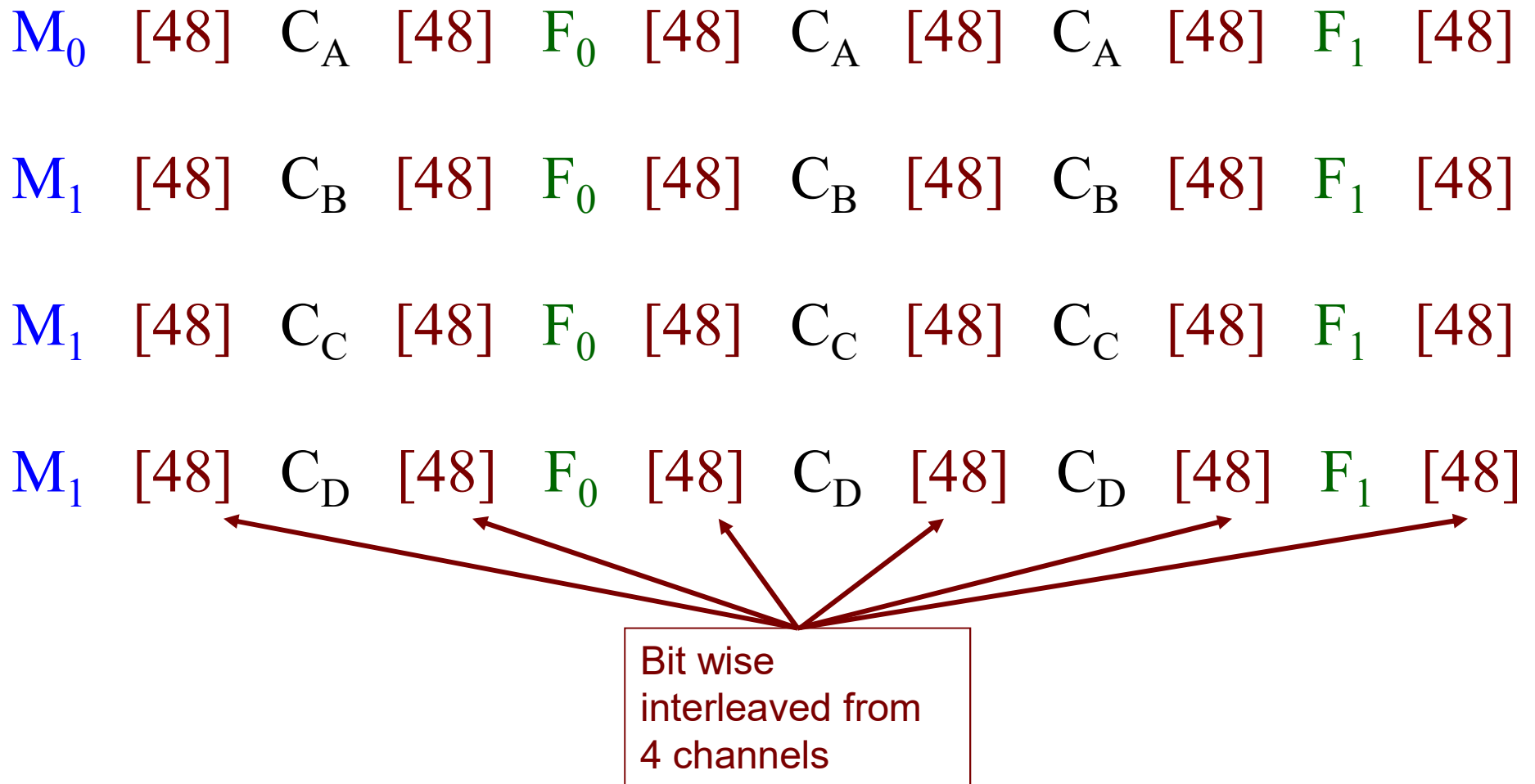
All [48]'s : Data bits, interleaved
from 4 channels

C_A, C_B, C_C, C_D : Stuffing bits

F_0, F_1 : Main framing bits

M_0, M_1 : Sub-framing bits

DM1/2 Multiplexer Frame Format



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M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

Total Data bits = $48 \times 6 \times 4 = 1152$ bits

Total overhead bits = $6 \times 4 = 24$ bits

Efficiency = $1152/1176 \approx 98\%$

DM1/2 Multiplexer Frame Format

M_0 [48] C_A [48] F_0 [48] C_A [48] C_A [48] F_1 [48]

M_1 [48] C_B [48] F_0 [48] C_B [48] C_B [48] F_1 [48]

M_1 [48] C_C [48] F_0 [48] C_C [48] C_C [48] F_1 [48]

M_1 [48] C_D [48] F_0 [48] C_D [48] C_D [48] F_1 [48]

$M_0, = F_0 = 0$

$M_1, = F_1 = 1$

Values of C_A, C_B, C_C, C_D , depend
on stuffing types

DM1/2 Multiplexer Frame Format

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Framing bits make a periodic pattern 010101... : used to synchronize on the frame

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$M_0M_1M_1M_1 = 0111$ is used to identify sub-frames.

DM1/2 Multiplexer Frame Format

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Framing bits make a periodic pattern 010101... : used to synchronize on the frame

$M_0M_1M_1M_1 = 0111$ is used to identify sub-frames.

Pattern $M_0M_1M_1M_1 = 0111$ also confirms genuine $F_0F_1F_0F_1$ pattern

Complication in Switching & Signaling

- All channels are NOT active always: some transmits, some are idle
- multiplexer's slots may be underutilized
- Additional input channels are allowed

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Complexity

- NO guaranty to hold: No. of active channels $\leq$ No. of available slots
- Complicated switching operations

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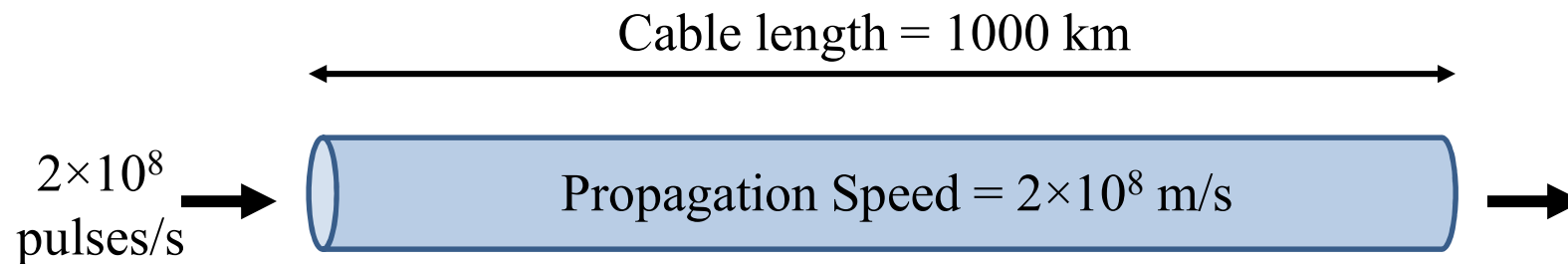
Management

- Use previous statistics to minimize the overload

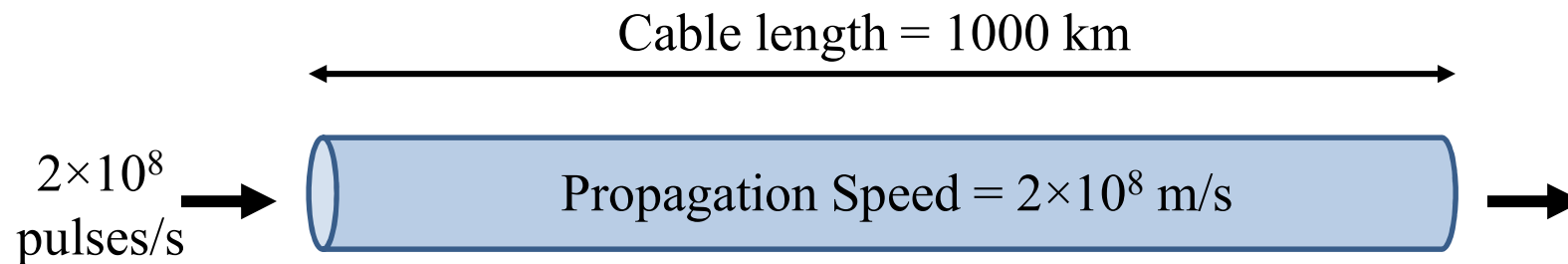
Asynchronous Channels & Bit Stuffing

- Synchronization is lost due to different reasons
 - variation in oscillation of oscillators
 - variation in temperature of carrying cables

Asynchronous Channels & Bit Stuffing

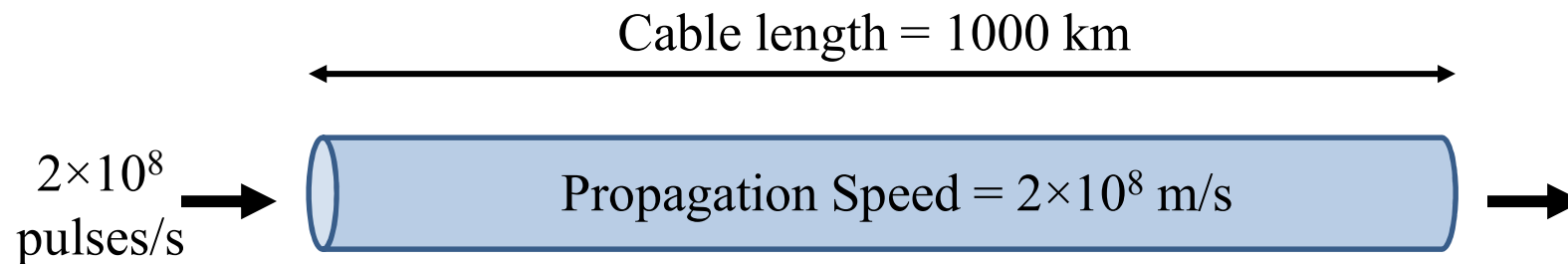


Asynchronous Channels & Bit Stuffing



Propagation time for a pulse through the cable = $1000 \text{ km} / (2 \times 10^8 \text{ m/s})$
= 1/200 second

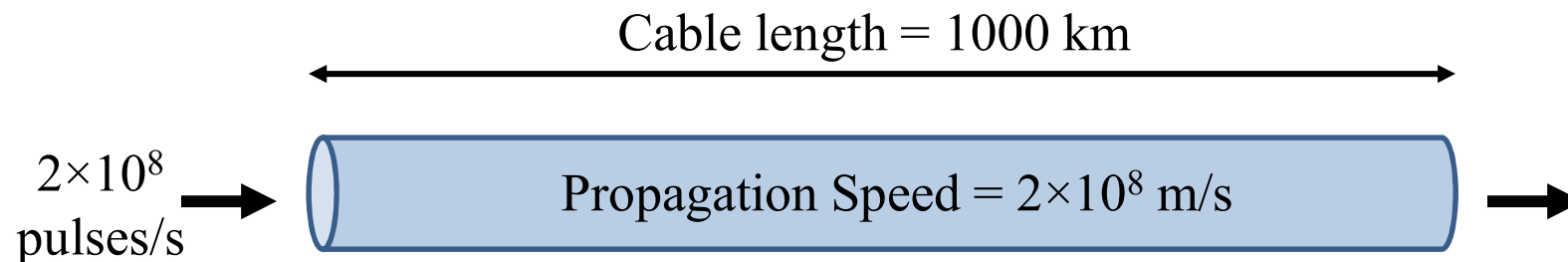
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No. of Pulses in transit within this time = $1/200 \text{ second} * (2 \times 10^8 \text{ pulses/s})$
= 10^6 pulses

Asynchronous Channels & Bit Stuffing

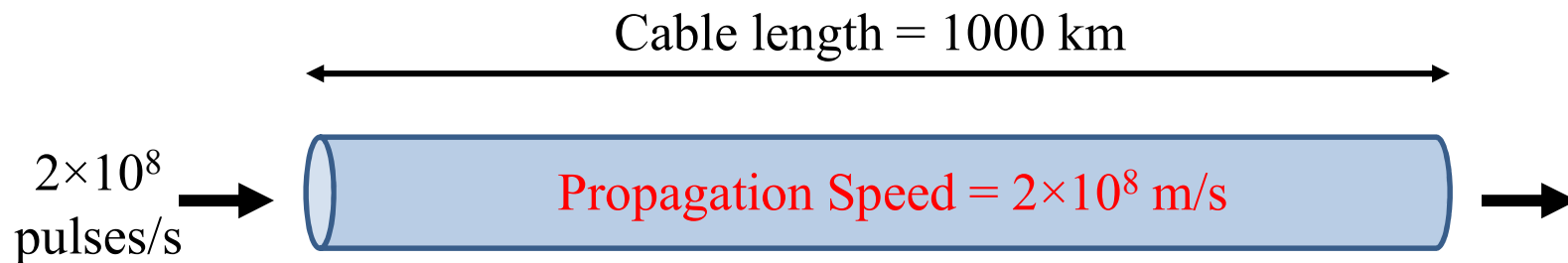


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1° F increase in cable temperature $\Rightarrow$ **0.01% increase in propagation speed**

Asynchronous Channels & Bit Stuffing

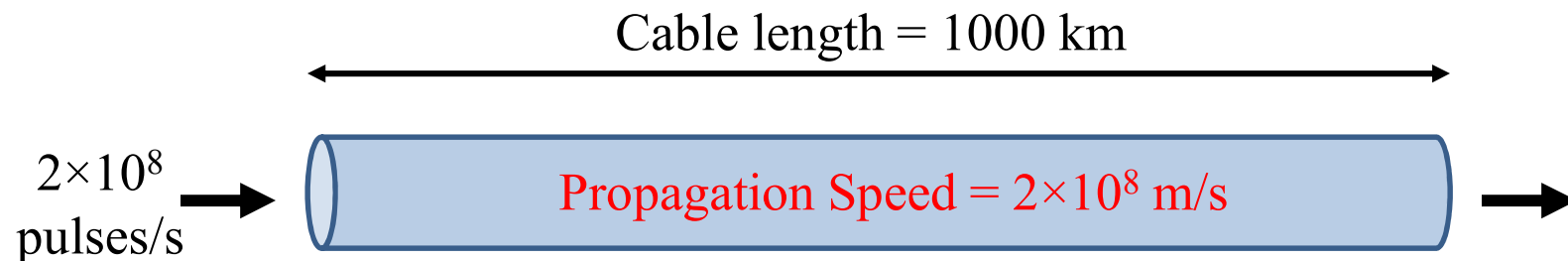


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Asynchronous Channels & Bit Stuffing



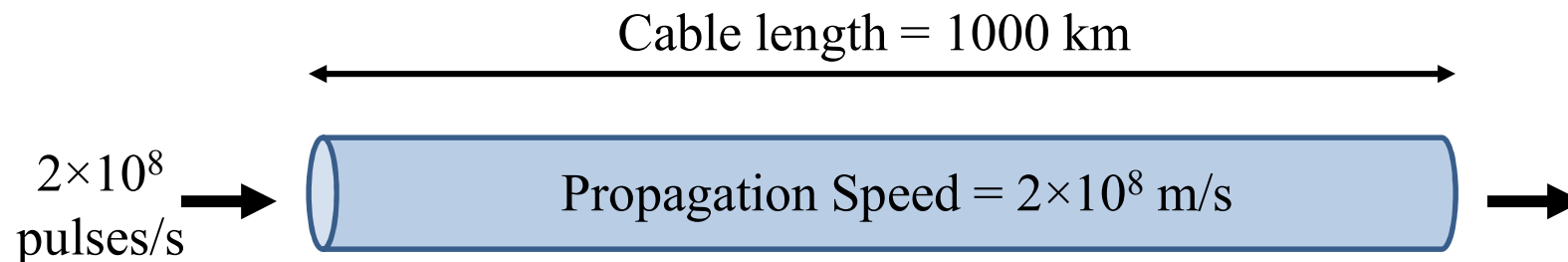
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1° F increase in cable temperature => 0.01% increase in propagation speed

Extra pulses must be stored somewhere

Asynchronous Channels & Bit Stuffing

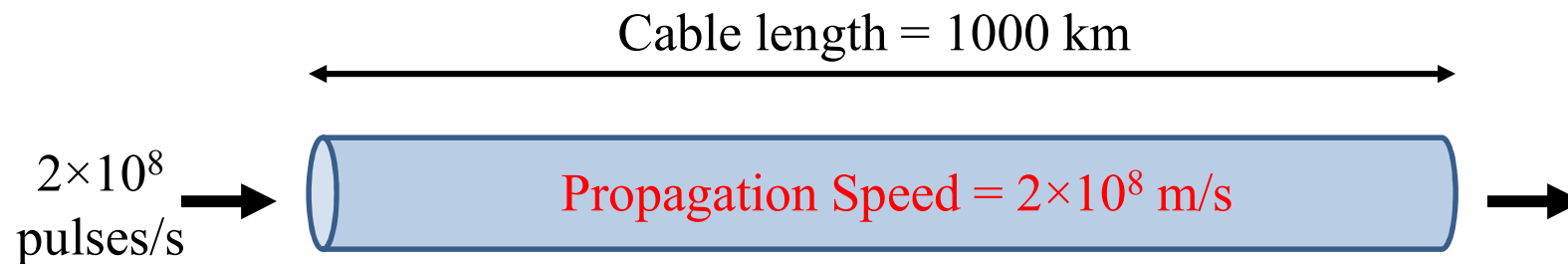


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drop in cable temperature $\Rightarrow$ rate of received pulses drops $\Rightarrow$ vacant slots in multiplexer $\Rightarrow$ vacant slots to be stuffed with dummy bits

Asynchronous Channels & Bit Stuffing



Findings:

- Even synchronous multiplexer system may receive bits asynchronously
- We need
 - storage for / removal of extra bits
 - Stuffing for vacant slots

Asynchronous Channels & Bit Stuffing

Stuffing classification

- Positive
- Negative
- Positive/negative

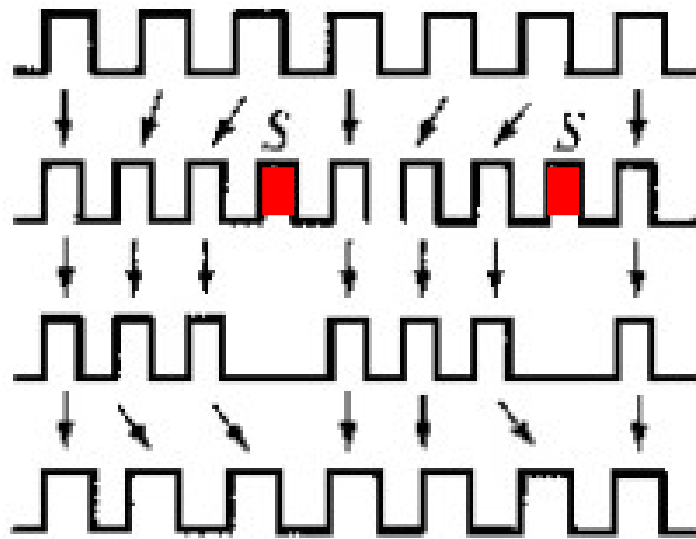
Asynchronous Channels & Bit Stuffing

Positive Stuffing

- Multiplexer's rate $>$ total max rates of all incoming channels
- Time slots in multiplexer remain vacant at some stage
- Pulse stuffing becomes necessary
- Stuffed bit and position are encoded using C-bits

Asynchronous Channels & Bit Stuffing

Positive Stuffing



Input signal to multiplexer

Transmitted signal with stuffing

Un-stuffed signal

Output signal with smoothing

Asynchronous Channels & Bit Stuffing

Negative Stuffing

- Multiplexer's rate $<$ total rates of all incoming channels
- Time slots in multiplexer becomes overloaded at some stage
- Multiplexer cannot accommodate all incoming signals
- Information of left-out pulses and position is transmitted through overhead bits

Asynchronous Channels & Bit Stuffing

Positive/Negative Stuffing

- Combination of positive and negative stuffing
- Nominal rate of multiplexer = total nominal rate of incoming channels
- positive stuffing at sometimes and negative at others
- All information is transmitted through overhead bits

Transmitting Stuffing Information through C-bits

M_0 [48] C_A [48] F_0 [48] C_A [48] C_A [48] F_1 [48]

M_1 [48] C_B [48] F_0 [48] C_B [48] C_B [48] F_1 [48]

M_1 [48] C_C [48] F_0 [48] C_C [48] C_C [48] F_1 [48]

M_1 [48] C_D [48] F_0 [48] C_D [48] C_D [48] F_1 [48]

- Four incoming channels: A, B, C, D
- Only 1 bit stuffing is allowed per channel per frame

Transmitting Stuffing Information through C-bits

M_0 [48] C_A [48] F_0 [48] C_A [48] C_A [48] F_1 [48]

M_1 [48] C_B [48] F_0 [48] C_B [48] C_B [48] F_1 [48]

M_1 [48] C_C [48] F_0 [48] C_C [48] C_C [48] F_1 [48]

M_1 [48] C_D [48] F_0 [48] C_D [48] C_D [48] F_1 [48]

Channel

Stuffing

No-Stuffing

A

$C_A C_A C_A = 111$

$C_A C_A C_A = 000$

B

$C_B C_B C_B = 111$

$C_B C_B C_B = 000$

C

$C_C C_C C_C = 111$

$C_C C_C C_C = 000$

D

$C_D C_D C_D = 111$

$C_D C_D C_D = 000$

Transmitting Stuffing Information through C-bits

M_0	[48]	C_A	[48]	F_0	[48]	C_A	[48]	C_A	[48]	F_1	[48]
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M_1	[48]	C_D	[48]	F_0	[48]	C_D	[48]	C_D	[48]	F_1	[48]

- The stuffed bit is the first bit following F_1 in the corresponding sub-frame